BRAIN 11

Semax

Synthetic 7-amino-acid peptide (Met-Glu-His-Phe-Pro-Gly-Pro), an analog of the ACTH 4-10 fragment



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Peptide	Stroke recovery, migraine	Small human studies	Not FDA approved · compounding allowed (nasal)

STUDIED FOR

Stroke recovery	Migraine, nerve pain	Brain networks
HUMAN + ANIMAL	HUMAN	HUMAN

Nerve growth factors	Blood clotting
ANIMAL	ANIMAL

Human data are limited. All human studies used nasal drops, mostly in Russia. FDA found no study of Semax given by injection.

EVIDENCE & WHAT'S KNOWN

EVIDENCE

SMALL HUMAN
LEVEL 3 OF 5

- Small, mostly uncontrolled Russian studies
- Nasal drops only; no injection studies
- No registered trial

NOT YET ESTABLISHED

- Pharmacokinetics in people
- Safety of injection
- Abuse potential
- Bleeding risk with other drugs

HOW IT WORKS IN THE BODY

Semax → BDNF and NGF → Nerve protection → Stroke recovery

- **ACTH 4-10 analog** without ACTH's hormone effects
- **In rats,** raised BDNF and NGF in the brain; smaller stroke damage
- **In mice,** boosted amphetamine-triggered dopamine release
- **FDA:** mechanisms poorly understood; not confirmed in people

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established in people	Nasal drops in all human studies
PHARMACOKINETICS	**SOLD AS**
Not established (no human data)	Compounded (Rx, nasal) or "research use only"
MONITORED IN TRIALS	**SPORT (WADA)**
Stroke rating scales, BDNF, pain scores	Not on the 2026 list

REGULATORY	FDA	HHS	WADA
	Not approved	Compounding allowed, nasal free base (Oct 2026)	Not named; S0 status not stated

REFERENCES

1. FDA, Pharmacy Compounding Advisory Committee briefing on Semax, Jul 2026
2. HHS, Federal Register notice 2026-22192, Oct 30, 2026
3. WADA, 2026 Prohibited List

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